Experiment 24: K-Nearest Neighbors (KNN) Classifier

Aim: To predict whether a patient has a medical condition using the KNN algorithm.

Algorithm:

1. Import the required libraries.
2. Create patient training data.
3. Train the KNN classifier.
4. Get patient features and K value from the user.
5. Predict and display the result.

Code:

```
from sklearn.neighbors import KNeighborsClassifier

X = [[20, 120], [25, 130], [30, 140],
    [45, 180], [50, 190], [55, 200]]
y = [0, 0, 0, 1, 1, 1]

k = int(input("Enter K: "))

model = KNeighborsClassifier(n_neighbors=k)
model.fit(X, y)

age = int(input("Enter age: "))
bp = int(input("Enter blood pressure: "))

result = model.predict([[age, bp]])

if result[0] == 1:
   print("Patient has the condition")
else:
   print("Patient does not have the condition")
```

Input:

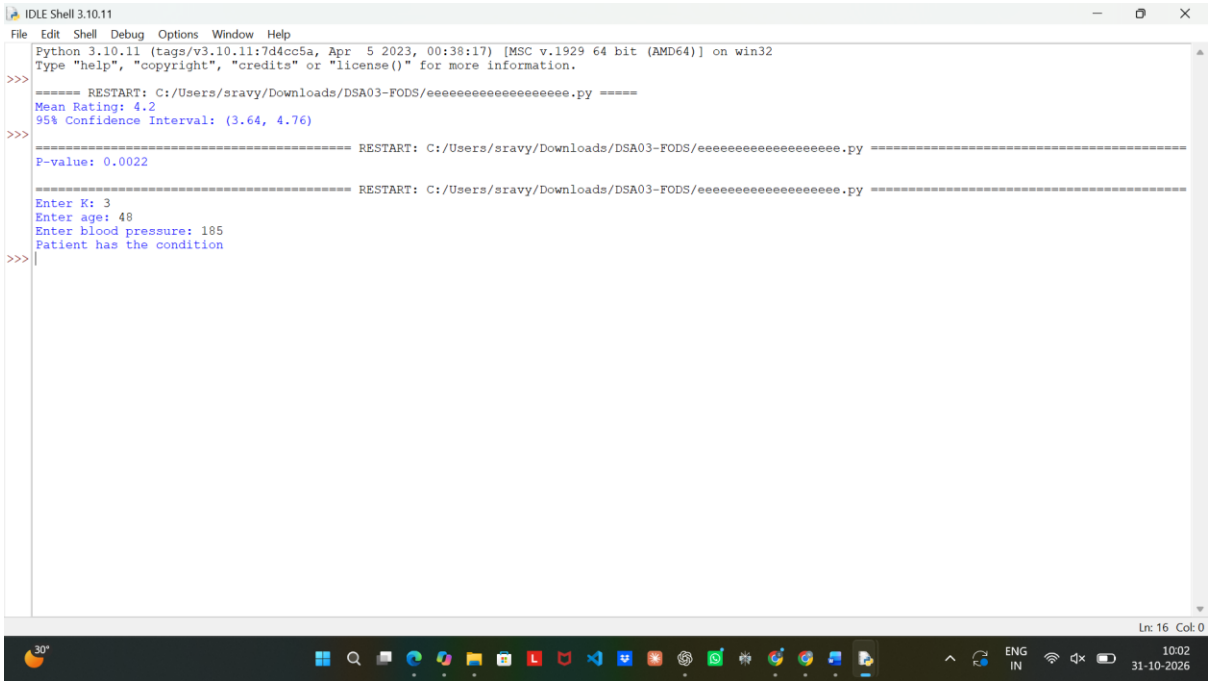
Enter K: 3

Enter age: 48

Enter blood pressure: 185

Output:

Patient has the condition



Result: The KNN classifier successfully predicted the patient's condition.